

Section 6B. Forward and Backward Stepwise Selection

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Forward Stepwise Selection

To overcome the computational challenges of Best Subset Selection, we can use *Forward Stepwise Selection*

Algorithm:

1. $\mathcal{M}_0$ denotes the null model $\hat{y} = \hat{\beta}_0$ with no inputs
2. For $k = 0$ to $p - 1$, do the following:
 - 2.1 Fit all $p - k$ models that augment the model $\mathcal{M}_k$ with one additional input
 - 2.2 Out of all those $p - k$ models, pick the one with the smallest (training) RSS, $\mathcal{M}_{k+1}$
3. End For-loop
4. Out of the p models $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$, find the model that minimizes the cross-validated **testing error**

Remark: This algorithm is not guaranteed to find the best possible model

Backward Stepwise Selection

Another alternative to overcome the computational burden of Best Subset Selection is to use the *Backward Stepwise Selection*

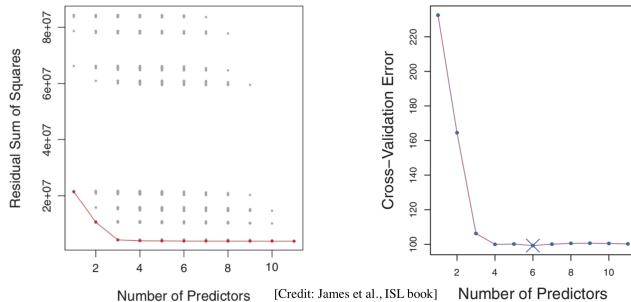
Algorithm:

1. $\mathcal{M}_p$ denotes the full model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 \cdots + \hat{\beta}_p x_p$, including all inputs
2. For $k = p, p-1, \dots, 1$, do the following:
 - 2.1 Fit all k models that contain all but one of the inputs in the model $\mathcal{M}_k$
 - 2.2 Out of all those k models, pick the one with the smallest RSS and call it $\mathcal{M}_{k-1}$
3. End For-loop
4. Out of the p models $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$, find the model that minimizes the cross-validated testing error

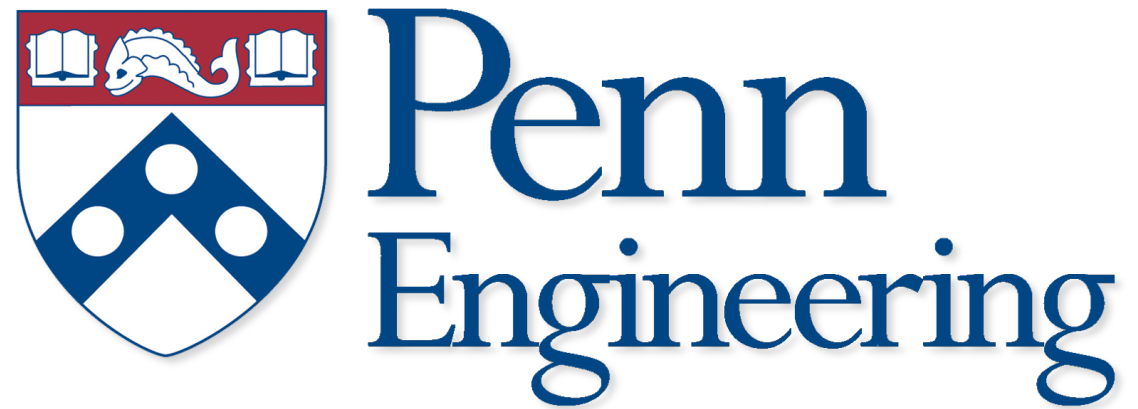
Remark: This algorithm is not guaranteed to find the best possible model

Model Selection: Example

Using the **Credit** dataset, we run the Best Subset Selection algorithm



- ▶ Left figure: RSS's for all models with k predictors, for $k = 1, \dots, 11$
- ▶ Right figure: Estimated testing errors using 10-fold CV for the best 11 models



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